MobilePixorNeXt + SimAM · BEV to grouped detection heads

Backbone

hist14 BEV
14 × 800 × 704

Conv3×3, 14→32, s=2
BN + SiLU

Stem /2
32 × 400 × 352

Conv3×3, 32→32, s=1
BN + SiLU

Downsample 2
Conv3×3, 32→48, s=2
BN + SiLU

NeXt block 1
48→120→48

NeXt block 2
48→120→48

NeXt block 3
48→120→48

Stage 2 / C3 + SimAM
48 × 200 × 176 /4

Residual SimAM
48 × 200 × 176
λ=0.0001; β₀=0.001

L3: Conv1×1
48→32

P3

G

Output Conv3×3, 32→32
BN + SiLU

Bilinear ↑2
Conv3×3, 48→32
BN + SiLU

P4

G

Bilinear ↑2
DWConv3×3, 48→48
BN + SiLU

Downsample 3
Conv3×3, 48→96, s=2
BN + SiLU

NeXt block 1
96→240→96

NeXt block 2
96→240→96

NeXt block 3
96→240→96

NeXt block 4
96→240→96

Focal Context
D=64; dilation 1,2,3
γ₀=0.001

L4: Conv1×1
96→48

Stage 3 / C4
96 × 100 × 88 /8

Downsample 4
Conv3×3, 96→128, s=2
BN + SiLU

NeXt block 1
128→320→128

NeXt block 2
128→320→128

L5: Conv1×1
128→48

Stage 4 / C5
128 × 50 × 44 /16

G: P = U + 2σ(DWConv(L + U)) ⊙ L

(B) Residual SimAM

λ = 0.0001 · β₀ = 0.001

Squared deviation d; spatial variance v
per batch / channel, FP32

e = d / [4(v + λ)] + 0.5

SimAM(X) = X ⊙ sigmoid(e)

Y = X + β [SimAM(X) − X]

(A) NeXt block

C = 48 / 96 / 128 · expansion 2.5

X

DW7×7
C→C

BN

PW1×1
C→2.5C

SiLU

PW1×1
2.5C→C

BN

γ

+

Y

Identity shortcut

Y = X + γ ⊙ F(X) · γ₀ = 10⁻⁵

Group 1: Car

K=1 · five independent prediction branches

Conv → Heatmap
1 × 200 × 176

Conv → Offset
2 × 200 × 176

Conv → Size
2 × 200 × 176

Conv → Yaw
2 × 200 × 176

Conv → IoU quality
1 × 200 × 176

Per branch: Conv3×3 → Conv3×3 → Conv1×1

Shared backbone features: 32 × 200 × 176

Group 2: Pedestrian / Cyclist

K=2 · five independent prediction branches

Conv → Heatmap
2 × 200 × 176

Conv → Offset
2 × 200 × 176

Conv → Size
2 × 200 × 176

Conv → Yaw
2 × 200 × 176

Conv → IoU quality
1 × 200 × 176

Per branch: Conv3×3 → Conv3×3 → Conv1×1

Shapes: C × H × W (batch omitted). G: scale-gated fusion. Cuboids are schematic; head outputs are raw predictions.